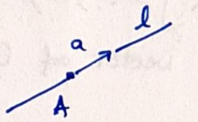
Seminar 2 - Warm-up

1.03.2026

2.1. Parametric equations for the line l.

a) l contains $A(1,2)$ and l is $\parallel$ to the vector $a(3,-1)$.

$$l: \begin{cases} x = 1+3t \\ y = 2-t \end{cases}, t \in \mathbb{R}$$



$$\Leftrightarrow l: \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} t, \ t \in \mathbb{R}.$$

$$\Leftrightarrow l: \ x = 1+3t, \quad y = 2-t, \ t \in \mathbb{R}.$$

b) l contains the origin and is $\parallel$ to $b(4,5)$.

$$l: \begin{cases} x = 0 + 4 \cdot s \\ y = 0 + 5 \cdot s \end{cases}, s \in \mathbb{R}$$

c) l contains $M(1,7)$ and is $\parallel Oy$.

$\parallel Oy \Rightarrow l$ has the same direction vectors as Oy

The basis of $\mathbb{V}^2$ is $(i,j) \Rightarrow j$ is a direction vector for $l \Rightarrow$

$\Rightarrow$ the components of j are $(0,1)$. $\Rightarrow l: \begin{cases} x = 1 \\ y = 7+t \end{cases}, t \in \mathbb{R}$?

2.2. For the lines in 2.1,

a) det. a Cartesian eq. for l.

$$l: \begin{cases} x = 1+3t \\ y = 2-t \end{cases}, t \in \mathbb{R} \Rightarrow \begin{cases} t = \frac{x-1}{3} \\ t = \frac{y-2}{-1} \end{cases} \Rightarrow l: \frac{x-1}{3} = \frac{y-2}{-1}$$

$$\Rightarrow l: -x+1 = 3y-6$$

$$l: x+3y-7=0$$

Since $a(3,-1)$ direction vector, all other direction vectors are:

$$\{\lambda \cdot a = (3\lambda, -\lambda) : \lambda \in \mathbb{R} \setminus \{0\}\}.$$

b) describe all direction vectors for l.

$$l: \begin{cases} x = 4s \\ y = 5s \end{cases}, s \in \mathbb{R} \Rightarrow \begin{cases} s = \frac{x}{4} \\ s = \frac{y}{5} \end{cases} \Rightarrow l: \frac{x}{4} = \frac{y}{5} \Rightarrow l: 5x-4y=0.$$

Direction vectors: $\{\lambda \cdot b(4,5) = (4\lambda, 5\lambda), \lambda \in \mathbb{R}$

c) $l: \begin{cases} x = 1 \\ y = 7 + t \end{cases}$, because $(x, y) = (1,7) + t\cdot(0,1) = (1,7) + (0,t) = (1, 7+t)$.
$t \in \mathbb{R}$

$\Rightarrow l: x = 1$ (y arbitrary)

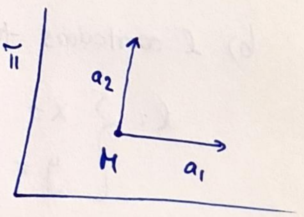
$\Rightarrow$ direction vectors: $\{\lambda \cdot j = (0,\lambda) : \lambda \in \mathbb{R}^*\}$.

(because the direction vector of Oy is $(0,1)$).

2.3. Parametric eq. for plane π:

a) π contains $M(1,0,2)$ and is $\parallel$ to $a_1(3,-1,1)$ and $a_2(0,3,1)$.

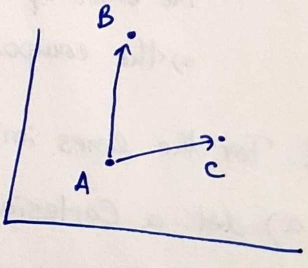
$\pi: \begin{cases} x = 1 + 3t + 0\cdot s \\ y = 0 - 1\cdot t + 3\cdot s \\ z = 2 + 1\cdot t + 1\cdot s \end{cases}$, $t, s \in \mathbb{R}$.



$\Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}\cdot t + \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}\cdot s$, $t, s \in \mathbb{R}$.

$\Leftrightarrow \pi:$ $x = 1 + 3t$, $y = -t + 3s$, $z = 2 + t + s$, $t, s \in \mathbb{R}$.

b) π contains $A(-2,1,1)$, $B(0,2,3)$, $C(1,0,-1)$.



$\vec{AB} = B(0,2,3) - A(-2,1,1) = (2,1,2) \Rightarrow \vec{AB} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

$\vec{AC} = C - A = (+3,-1,-2) \Rightarrow \vec{AC} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$

$\vec{AB}, \vec{AC}$ are vectors represented with points in π.

$\vec{AB}, \vec{AC}$ lin. indep. $\Rightarrow$ they form a basis of the direction space $D(\pi)$.

$\pi: \begin{cases} x = -2 + 2t + 3s \\ y = 1 + t - s \\ z = 1 + 2t - 2s \end{cases}$, $t, s \in \mathbb{R}$

A. (fixed) $\quad \vec{AB} \quad \vec{AC}$

$\Leftrightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}\cdot t + \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}\cdot s$, $t, s \in \mathbb{R}$.

c) π contains the point $A(1,2,1)$ and is $\parallel$ to i and j.

$i=\begin{bmatrix}1\\0\\0\end{bmatrix}$, $j=\begin{bmatrix}0\\1\\0\end{bmatrix}$ since, by convention, (i,j,k) is a basis of V^3.

$$\pi : \begin{cases} x = \underset{\downarrow A.}{1} + t\cdot 1 + s\cdot 0 \\ y = 2 + t\cdot 0 + s\cdot 1 \\ z = 1 + t\cdot 0 + s\cdot 0 \end{cases} \iff \pi : \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}1\\2\\1\end{bmatrix} + t\cdot\begin{bmatrix}1\\0\\0\end{bmatrix} + s\cdot\begin{bmatrix}0\\1\\0\end{bmatrix},$$

$t, s \in \mathbb{R}$

d) π contains $M(1,7,1)$ and is $\parallel$ to the coordinate plane Oyz.

- a basis for $D(Oyz)$ is (j,k).

$$j=\begin{bmatrix}0\\1\\0\end{bmatrix},\quad k=\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$\pi : \begin{cases} x = 1 \\ y = 7+t \\ z = 1+s \end{cases}, t,s \in \mathbb{R}. \iff \pi : \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}1\\7\\1\end{bmatrix} + t\cdot\begin{bmatrix}0\\1\\0\end{bmatrix} + s\cdot\begin{bmatrix}0\\0\\1\end{bmatrix}$$

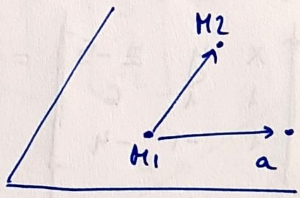
$t, s \in \mathbb{R}$.

e) π contains $M_1(5,3,4)$, $M_2(1,0,1)$ and is $\parallel$ to $a(1,3,-3)$.

$\overrightarrow{M_2M_1}(5-1, 3-0, 4-1) = \overrightarrow{M_2M_1}(4,3,3)$ is a vector represented with points in $\pi \Rightarrow \pi \parallel \overrightarrow{M_2M_1}$.

vectors $\overrightarrow{M_2M_1}$ and a are lin. ind. vectors $\Rightarrow$

$\Rightarrow$ they form a basis of $D(\pi)$.



$$\pi : \begin{cases} x = 5 + 4t + s \\ y = 3 + 3t + 3\cdot s \\ z = \underset{\downarrow M_1}{4} + \underset{\downarrow \overrightarrow{M_2M_1}}{3t} + \underset{\downarrow a}{(-3)\cdot s} \end{cases}, t,s \in \mathbb{R}$$

$$\iff \pi : \begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}5\\3\\4\end{bmatrix} + \begin{bmatrix}4\\3\\3\end{bmatrix}\cdot t + \begin{bmatrix}1\\3\\-3\end{bmatrix}\cdot s, \quad t,s \in \mathbb{R}.$$

2.4. Det. Cartesian eq. for the plane π:

a) π : $x = 2+3u-4v$, $y = 4-v$, $z = 2+3u$, $u,v \in \mathbb{R}$.

$$\pi: \begin{cases} x = 2+3u-4v \\ y = 4-v \\ z = 2+3u \end{cases}, \quad u,v \in \mathbb{R}$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} + 3\cdot\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\cdot u - \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} v, \quad u,v \in \mathbb{R}$$

$$\Rightarrow \begin{bmatrix} x-2 \\ y-4 \\ z-2 \end{bmatrix} = \underset{a}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}\cdot 3u - \underset{b}{\begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}}\cdot v, \quad u,v \in \mathbb{R}.$$

$$\pi: \begin{array}{c} \\ a \rightarrow \\ b \rightarrow \end{array} \begin{vmatrix} x-2 & y-4 & z-2 \\ 1 & 0 & 1 \\ 4 & 1 & 0 \end{vmatrix} = 0$$

$$\Leftrightarrow (x-2)(-1) - (y-4)(-4) + (z-2) = 0$$

$$\Rightarrow \pi: -x + 4y + z - 16 = 0.$$

b) $$\pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix}\cdot u + \begin{bmatrix} 1 \\ -1 \\ -4 \end{bmatrix} v, \quad u,v \in \mathbb{R}.$$

$$\Rightarrow \pi: \begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = 0$$

$$\Leftrightarrow \cancel{x}\ \pi: 2x + 10y - z\cdot 2 + 10 = 0 \ |:2$$

$$\Leftrightarrow \pi: x + 5y - z + 5 = 0$$

Compute determinant M1: $$x\cdot\underbrace{\begin{vmatrix} 1 & 6 \\ -1 & -4 \end{vmatrix}}_{2} - y\cdot\underbrace{\begin{vmatrix} 1 & 6 \\ 1 & -4 \end{vmatrix}}_{-10} + (z-5)\cdot\underbrace{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}}_{-2} = 0$$

Compute determinant M2:

$$\begin{vmatrix} x & y & z-5 \\ 1 & 1 & 6 \\ 1 & -1 & -4 \end{vmatrix} = \underline{-4x} + \cancel{5} 5 - z + \underline{6y + 4y} + \underline{6x} + 5 - z$$
$$= 2x + 10y + 10 - 2z$$

$$\begin{array}{ccc} x & y & z-5 \\ 1 & 1 & 6 \end{array}$$

2.5. Det. parametric eq. for the plane π:

a) $\pi: 3\underline{x} - 6\underline{y} + z = 0$

Choose x and y as parameters : $\begin{cases} x = x \\ y = y \\ z = -3x + 6y. \end{cases}$

$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix} \quad t, s \in \mathbb{R}.$

we „relabel" x and y.

b) $\pi: 2x - y - z - 3 = 0.$

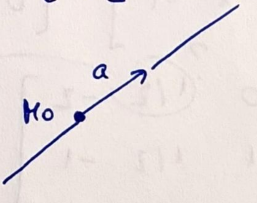
Choose $x = t$ and $z = s$ as parameters : $\begin{cases} x = t \\ y = 2x - z - 3 = 2t - s - 3. \\ z = s \end{cases}$

$\Rightarrow \pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix}}_{\substack{\text{ct.} \\ \text{(point)}}} + t \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + s \cdot \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

2.6. Det. parametric eq. for the line l:

a) l contains $M_0(2,0,3)$ and is $\parallel$ to $a(3,-2,-2)$.

$l: \begin{cases} x = 2 + 3t \\ y = 0 - 2t \\ z = 3 - 2t \end{cases}, t \in \mathbb{R} \iff l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \\ -2 \end{bmatrix} \cdot t$



b) l contains $A(1,2,3)$ and is $\parallel$ to $\underline{Oz \text{ axis}}$.

↳ k is a direction vector, $k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

$l: \begin{cases} x = 1 \\ y = 2 \\ z = 3 + 1 \cdot t \end{cases}, t \in \mathbb{R} \iff l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \cdot t$

c) l contains $M_1(1,2,3)$ and $M_2(4,4,4)$.

- a direction vector is $\overrightarrow{M_1M_2}$ or $\overrightarrow{M_2M_1}$
- a point on l is M_1 or M_2.

2.7. Cartesian eq. for the lines l in 2.6.

a) $l: \begin{cases} x=2+3t & \Rightarrow t=\frac{x-2}{3} \\ y=-2t & \Rightarrow t=-\frac{y}{2} \\ z=3-2t & \Rightarrow t=\frac{z-3}{-2} \end{cases} \Rightarrow \frac{x-2}{3}=\frac{y}{-2}=\frac{z-3}{-2}$

Choose 2 equations: $l: \begin{cases} \frac{x-2}{3}=\frac{y}{-2} \\ \frac{y}{-2}=\frac{z-3}{-2} \end{cases} \Leftrightarrow l: \begin{cases} -2x+4=3y \\ -2y=-2z+6 \end{cases} \Leftrightarrow$

$\Leftrightarrow l: \begin{cases} 2x+3y-4=0 \\ 2y-2z+6=0 \end{cases}$

b) $l: \begin{cases} x=1 \\ y=2 \\ z=3+t \end{cases}$ $\rightarrow$ z is arbitrary ?! $\Rightarrow l: \begin{cases} x=1 \\ y=2 \end{cases} \Leftrightarrow l: \begin{cases} x-1=0 \\ y-2=0 \end{cases}$

2.8. Parametric eq. for the line contained in the planes $x+y+2z-3=0$ and $x-y+z-1=0$.

$l: \begin{cases} x+y+2z-3=0 \\ x-y+z-1=0 \end{cases}$ (*)

$2-\frac{1}{2}$

$-3+1$

$\begin{bmatrix} 1 & 1 & 2 & -3 \\ 1 & -1 & 1 & -1 \end{bmatrix} \overset{L_2-L_1\to L_2}{\sim} \begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & -2 & -1 & 2 \end{bmatrix} \xrightarrow{-\frac{1}{2}\cdot L_2\to L_2} \begin{bmatrix} 1 & 1 & 2 & -3 \\ 0 & 1 & 1/2 & -1 \end{bmatrix} \xrightarrow{L_1-L_2\to L_1}$

$\sim \begin{bmatrix} 1 & 0 & \overset{?}{1/2} & -2 \\ 0 & 1 & 1/2 & -1 \end{bmatrix}$ $\rightarrow$ $3/2$?

$(*) \Leftrightarrow \begin{cases} x+\frac{3z}{2}-2=0 \\ y+\frac{z}{2}-1=0 \end{cases}$ $\rightarrow$ why ?

if we choose the parameter $t=z/2$, then:

$\begin{cases} x=2-3t \\ y=1-t \end{cases} \Leftrightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t\cdot \begin{bmatrix} -3 \\ -1 \\ 0 \end{bmatrix}$ 2?

?.9. Cartesian eq. for the plane containing point Q and line ℓ.

a) $Q = (3,3,1)$, ℓ: $x = 2+3t$, $y = 5+t$, $z = 1+7t$, $t \in \mathbb{R}$.

$$Q = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}, \quad \ell: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}}_{= A} + t \cdot \underbrace{\begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix}}_{= v}, \quad t \in \mathbb{R}$$

$\Rightarrow$ the plane contains points Q and A
$\parallel v$.

$$\vec{QA} = A - Q = (2,5,1) - (3,3,1) = (-1,2,0)$$

$\Rightarrow \vec{QA}(-1,2,0)$. $\Rightarrow \pi \parallel \vec{QA}$.

$\vec{QA}$, v are lin. ind. vectors $\Rightarrow$ they form a basis of $D(\pi)$.

$$\pi: \begin{cases} x = -1 + (-1)\cdot t + 3\cdot s \\ y = 2 + 2\cdot t + 1\cdot s \\ z = 0 + 0\cdot t + 7\cdot s \end{cases} \quad t, s \in \mathbb{R}$$

?

$$\downarrow \; Q \qquad \downarrow \; \vec{QA} \qquad \downarrow \; v$$

$$\pi: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \cdot t + \begin{bmatrix} 3 \\ 1 \\ 7 \end{bmatrix} \cdot s \quad t, s \in \mathbb{R}.$$